

CS 6320 — Assignment 6

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Course: Deep Learning (CS 6320)

Part A — Portfolio tabular representation and model comparison (BGG)

Work type: Native tabular portfolio data (not proxy, not fallback case study).

Portfolio problem and representation

Item	Lock
Stakeholder	Hobby retailer screening preorder / initial stock for BGG-aware customers
Dataset	Kaggle <i>Board Games Database from BoardGameGeek</i> (<code>games.csv</code> , 21,925 rated games)
Unit	One row per game (BGGId)
Target	<code>high_rating = 1</code> if <code>AvgRating >= 7.0</code> , else <code>0</code> (~26.9% positive)
Representation	Native tabular — numeric design metadata, <code>GameWeight</code> , category flags, simple text-length features, parsed <code>goodplayers_count</code>
Part A path	Portfolio table directly (no Week 6 proxy translator)
Seed	6320
Implementation	6320-hw6/scripts/; end-to-end: <code>bash run_local.sh</code>

Features follow the Assignment 4 charter and Assignment 5 manifest: rating fields, rank columns, popularity counts, and mis-timed demand fields (`NumOwned`, `NumWant`, `NumWish`) are **hard-excluded**. Raw Name / Description text are not fed to models — only length/count proxies (same v1 charter).

Preprocessing and prediction-time rules

1. **Split first** — stratified random 70% / 15% / 15% by game row (unchanged from Assignment 5).
2. **Imputation** — numeric medians computed on **training rows only**, applied to validation/test (via `build_feature_matrix` on each split frame).
3. **Scaling** — `StandardScaler` fit on **training** features for logistic regression and MLP; trees use median-imputed raw features (scale-invariant).
4. **Categories** — eight `Cat:*` flags kept as 0/1 integers (low cardinality).
5. **Leakage** — no rating-derived or post-hoc demand columns in X ; each game appears in one split only.

Embeddings: Not used. Category fields are binary flags, not repeated high-cardinality IDs. Numeric fields are not identity codes. Sentence for writeup: *“Embeddings were not appropriate because categories are low-cardinality one-hot flags and there are no stable high-cardinality entity IDs; I used scaled numerics plus binary category columns instead.”*

Split and audit evidence

Split	Rows	Positive rate
train	15,347	26.9%
validation	3,289	27.7%
test	3,289	26.2%

Positive rates are stable across splits (within ~1.5 pp). Audit tables: `prep/bgg_split/split_audit/` (counts, numeric distributions, category positive rates).

Models compared (same target, split, metrics)

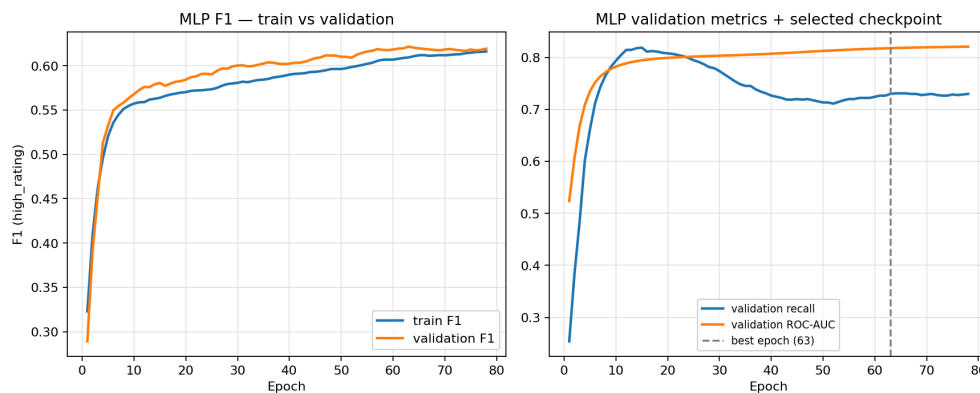
Model	Role	Key settings
Majority class	Sanity	Always predict not-high
Logistic regression	Simple linear baseline	L2, unweighted; train-fitted <code>StandardScaler</code>
Gradient boosted trees	Strong tabular benchmark	<code>HistGradientBoostingClassifier</code> , depth 6, lr 0.08, 300 trees
MLP	Neural tabular	64→32, ReLU, dropout 0.2, Adam (wd=1e-4), BCE pos_weight for imbalance, early stop on validation F1 (patience 15; best epoch 63)

Model selection uses **validation F1** on `high_rating=1`. Test metrics reported once for final comparison. No hyperparameter search beyond the fixed configs above.

Results table

Model	Val F1	Val recall	Val ROC-AUC	Test F1	Test recall	Test ROC-AUC
Majority class	0.000	0.000	0.500	0.000	0.000	0.500
Logistic regression	0.527	0.422	0.815	0.544	0.454	0.832
Gradient boosted trees	0.685	0.624	0.895	0.714	0.661	0.910
MLP (early stop, epoch 63)	0.621	0.730	0.818	0.619	0.747	0.830

Validation precision: logistic **0.703**, GBT **0.759**, MLP **0.541**. Test accuracy: logistic **0.801**, GBT **0.861**, MLP **0.759**.



MLP training curves — validation checkpoint selection

Figure 1: MLP train/validation F1 and validation recall/ROC-AUC; dashed line = selected checkpoint (epoch 63).

Model-choice recommendation

Tabular modeling is justified for this portfolio problem: structured metadata and category flags carry signal well above chance (ROC-AUC ~0.91 test for GBT vs 0.50 majority).

Recommended staged model: gradient boosted trees, not the MLP.

- GBT wins on **validation and test F1** and **ROC-AUC** with the best precision–recall balance for a screening task.
- The MLP reaches **higher recall** (~0.75 test) but **lower F1 and precision** than GBT — it over-predicts positives, which wastes buyer attention (false stock flags).
- Logistic regression remains a useful **transparent floor** but under-recalls hits (~0.45 test recall), consistent with Assignment 5.

- Neural tabular complexity (checkpoint tuning, opaque weights, calibration risk from Assignment 5) is **not justified** when GBT beats the MLP on the primary ranking metrics under the same split.

Change trigger: If a time-based split collapses GBT advantage, or if recalibrated probabilities become a hard requirement, revisit logistic + Platt scaling before adding neural capacity.

Practical constraints

Constraint	Implication
Interpretability	Logistic and tree splits easier to explain to a buyer than MLP weights
Cost / ops	All models train in seconds on local CPU (~22k rows); GBT and MLP similar inference cost
Maintainability	Fixed sklearn/torch pipelines; manifest documents feature exclusions
Data size	Tabular row count is modest — no need for large-scale neural capacity
Monitoring	Track slice metrics (e.g. light vs heavy games from A5) if GBT is deployed internally

Responsible-use concern

BGG ratings reflect **enthusiast community taste**, not universal quality. Assignment 5 slices showed **weak performance on light games** (low positive rate, low recall). A tabular model — especially one tuned for recall like the MLP — can still **over-flag** niche titles. Outputs should remain **human-review screening aids**, not automated stock decisions; sensitive-proxy risk from title text is mitigated in v1 by using length features only, but franchise memorization should be audited before client-facing use.

Run evidence (local)

- Command: `bash run_local.sh` from repo root
- Data: `6320-hw2/part_b/data/bgg/games.csv` (or `GAMES_CSV`)
- Artifacts: `outputs/comparison_table.csv`, `outputs/*/summary.json`, `outputs/mlp_early_stop/mlp_history.csv`, `outputs/plots/mlp_training_curves.png`, `logs/run_local.log`
- Compute: local CPU only; CHPC not required

Part B — Portfolio checkpoint and model-choice note

Current data readiness

Usable. BGG `games.csv` is local, parsed, and split under the Assignment 4/5 charter (stratified hold-out, seed 6320). Feature manifest and split audits are reproducible via `prepare_bgg_data.py` and `run_split_audit.py`. No proxy or fallback path was needed.

Current baseline / model status

Stage	Status
Majority + logistic baseline	Complete (A5); logistic metrics reproduced in A6 comparison
Assignment 5 evaluation	Complete — imbalance, calibration gaps, slice weakness documented
Gradient boosted trees	Trained (A6) — current best tabular candidate
MLP tabular	Trained (A6) — competitive recall but secondary to GBT on F1/ROC-AUC
Time-based split	Not yet run — still open from charter
Text / image modeling	Not started — Week 7+ relevance TBD

Next planned experiment

Time-based split by YearPublished (recent-release validation cohort) with the **same feature manifest and GBT config**, plus **recalibration** (Platt or isotonic on validation, checked on test). This tests the charter’s pre-release prediction story beyond random stratified hold-out.

Expected staged improvement

Honest pre-release metrics (likely lower than random split), better-calibrated probabilities for threshold decisions, and slice tables by release era. Success = documented trade-off between recall and precision under a decision-relevant split, not a single accuracy number.

How Week 6 evidence affects the final model-choice argument

Week 6 supports **keeping tabular methods central** to the portfolio: GBT on metadata alone reaches **test F1 0.71 / ROC-AUC 0.91**. It does **not** support elevating a neural tabular model — the MLP adds complexity without winning the comparison. The final presentation can argue: *structured features are sufficient for internal screening; modality-specific models (text embeddings, cover images) must beat this GBT protocol under a time split before they earn a place in the recommendation.*

Assignment 4 audit / charter updates

A4 item	Week 6 evidence	Update
Class imbalance	Logistic under-recalls; MLP boosts recall at precision cost; GBT balances best	Confirmed — metric choice must include F1/recall
Leakage exclusions	Same manifest; GBT gain is not from rating columns	Reinforced
GBT as initial candidate	Now trained; beats logistic and MLP on F1/AUC	Success criterion met for classical stage
Time-split generalization	Still random split only	Still untested — next experiment
Calibration / deployment	Not re-run in A6; GBT probabilities still need recalibration	Still open
Representativeness (light games)	Not re-sliced in A6	Still flagged from A5

Tabular relevance for the portfolio

Directly relevant. The client problem is inherently tabular metadata prediction. Embeddings are **not relevant** in v1 (low-cardinality categories). Simpler baselines remain **indirectly relevant** as floors and explainability anchors even if GBT becomes the primary candidate.

AI disclosure

Tool: Cursor (Composer)

AI-assisted: Repo scaffolding, model-comparison scripts, PyTorch MLP training loop, first-draft writeup prose.

My work: Locked native-tabular path; chose model set and fixed configs; ran and verified all metrics against saved outputs; model-choice recommendation based on comparison table.

Certification: I certify that all work not described above is my own, and I verified AI-generated content for accuracy.